

THE UNIVERSITY OF FAISALABAD

Department of Artificial Intelligence
Bachelor of Science in Artificial Intelligence

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

YouTube Comments Sentiment Analysis System

Using RoBERTa Transformer Model with Interactive Streamlit Dashboard

Document Version:	1.0
Date:	May 2026
Evaluator:	Mr. Muhammad Arham

Team Members

#	Name	Registration No.
1	Ali Hassan	2023-BS-AI-038
2	Hanzla Bhatti	2023-BS-AI-046
3	Waleed Amjad	2023-BS-AI-054
4	Taha Rehman	2023-BS-AI-090

Table of Contents

- 1. Introduction..... 4
 - 1.1 Purpose..... 4
 - 1.2 Scope..... 4
 - 1.3 Definitions, Acronyms, and Abbreviations..... 4
 - 1.4 References..... 5
 - 1.5 Overview..... 5
- 2. Overall Description..... 6
 - 2.1 Product Perspective..... 6
 - 2.2 Product Features..... 6
 - 2.3 User Classes and Characteristics..... 6
 - 2.4 Operating Environment..... 6
 - 2.5 Design and Implementation Constraints 7
 - 2.6 Assumptions and Dependencies..... 7
- 3. Specific Functional Requirements 8
 - 3.1 Data Loading and Management 8
 - FR-1.1: Dataset Loading 8
 - FR-1.2: Sample Configuration..... 8
 - FR-1.3: Data Caching 8
 - 3.2 Text Preprocessing..... 8
 - FR-2.1: Text Cleaning 8
 - FR-2.2: Processing Time Tracking 8
 - 3.3 Sentiment Classification 8
 - FR-3.1: Model Loading..... 8
 - FR-3.2: Batch Inference..... 8
 - FR-3.3: Label Mapping..... 8
 - FR-3.4: Model Evaluation Toggle 8
 - 3.4 KPI Dashboard..... 9
 - FR-4.1: Summary Metrics..... 9
 - FR-4.2: Model Performance Metrics 9
 - 3.5 Keyword Analysis and Visualization..... 9
 - FR-5.1: Overall Keyword Frequency..... 9
 - FR-5.2: Per-Sentiment Keyword Frequency..... 9
 - FR-5.3: Word Cloud Generation..... 9
 - 3.6 Live Prediction..... 9
 - FR-6.1: User Input Interface 9

2

FR-6.2: Real-Time Classification	9
FR-6.3: Spinner Feedback.....	10
4. Non-Functional Requirements	11
4.1 Performance Requirements	11
4.2 Reliability and Availability	11
4.3 Usability Requirements.....	11
4.4 Security Requirements	11
4.5 Maintainability Requirements.....	11
4.6 Portability Requirements	12
5. System Architecture and Models	13
5.1 System Architecture Overview	13
5.2 Technology Stack.....	13
5.3 Data Flow.....	13
5.4 Module Descriptions.....	13
5.4.1 app.py — Main Application Module	13
5.4.2 utils.py — Utility Module.....	13
6. Constraints and Limitations	15
6.1 Hardware Constraints.....	15
6.2 Software Constraints.....	15
6.3 Academic Scope Limitations	15
Appendix A: Requirements Traceability Matrix.....	16
Appendix B: Revision History.....	16

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for the YouTube Comments Sentiment Analysis System. The system leverages state-of-the-art transformer-based deep learning models to classify YouTube comments into Positive, Negative, and Neutral sentiment categories. This document is intended for developers, academic evaluators, and stakeholders involved in the design, development, and assessment of the system.

1.2 Scope

The YouTube Comments Sentiment Analysis System, hereafter referred to as "YT-Sentiment," is a web-based application that performs end-to-end sentiment classification on real-world YouTube comment datasets. The system scope encompasses:

- Automated data preprocessing and text cleaning pipeline using NLTK.
- Sentiment classification using the RoBERTa transformer model (cardiffnlp/twitter-roberta-base-sentiment-latest) from Hugging Face.
- An interactive Streamlit dashboard displaying key performance indicators (KPIs).
- Live sentiment prediction functionality for user-provided comment input.
- Visual analytics including word clouds and frequency bar charts per sentiment class.
- GPU-accelerated inference using PyTorch with CUDA support.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
NLP	Natural Language Processing — computational techniques for human language analysis
RoBERTa	Robustly Optimized BERT Pretraining Approach — a transformer language model
BERT	Bidirectional Encoder Representations from Transformers
GPU	Graphics Processing Unit — used for accelerated deep learning inference
CUDA	Compute Unified Device Architecture — NVIDIA parallel computing platform
API	Application Programming Interface
KPI	Key Performance Indicator
NLTK	Natural Language Toolkit — Python library for text preprocessing
UI	User Interface
F1-Score	Harmonic mean of Precision and Recall — model evaluation metric

1.4 References

- Devlin, J. et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv:1810.04805.
- Liu, Y. et al. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692.
- Barbieri, F. et al. (2022). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. HuggingFace Model Hub.
- Streamlit Documentation. <https://docs.streamlit.io>
- YouTube Comments Sentiment Dataset by Amaan Poonawala. Kaggle.
- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

1.5 Overview

The remainder of this document is organized as follows: Section 2 provides an overall description of the system including product perspective, features, and user characteristics. Section 3 details specific functional requirements. Section 4 defines non-functional requirements. Section 5 covers system models and architecture. Section 6 lists constraints and assumptions.

2. Overall Description

2.1 Product Perspective

The YouTube Comments Sentiment Analysis System is a standalone academic project developed as part of the BS Artificial Intelligence program at The University of Faisalabad. It is designed as a self-contained web application that does not depend on external enterprise systems. The application interacts with the following external components:

- Hugging Face Transformers Hub: Source of the pre-trained RoBERTa model (cardiffnlp/twitter-roberta-base-sentiment-latest).
- YouTube Comments Sentiment Dataset (Kaggle): Provides labeled training and evaluation data containing real-world YouTube comments.
- PyTorch / CUDA Runtime: Provides the underlying GPU computation framework for model inference.
- NLTK: Supplies tokenization and stop-word removal capabilities for text preprocessing.

2.2 Product Features

The following are the primary features of YT-Sentiment:

1. Automated preprocessing pipeline (tokenization, stop-word removal, text normalization).
2. RoBERTa-based three-class sentiment classification (Positive, Negative, Neutral).
3. Configurable sample size slider for analysis (2,000 to 40,000 comments).
4. KPI dashboard displaying total analyzed comments, sentiment distribution percentages, model accuracy, precision, recall, and F1-score.
5. Per-class and overall keyword frequency charts.
6. Interactive word cloud visualization per sentiment class.
7. Live prediction interface for user-supplied YouTube comments.
8. GPU/CPU detection and dynamic device allocation.

2.3 User Classes and Characteristics

User Class	Description	Technical Level
Academic Evaluator	Faculty member assessing project quality and functionality	Moderate — familiar with ML concepts
Developer / Researcher	Team members extending or modifying the codebase	High — proficient in Python, ML, NLP
End User	Anyone analyzing YouTube comment sentiment via the dashboard	Low to Moderate — no coding required

2.4 Operating Environment

The system is designed to operate in the following environments:

- Operating System: Windows 10/11 or Linux (Ubuntu 20.04+).

- Hardware: Minimum 8 GB RAM; NVIDIA GPU with 4 GB VRAM recommended (CUDA 11.8 compatible) for accelerated inference; CPU fallback is supported.
- Python Version: 3.9 or higher.
- Web Browser: Any modern browser (Chrome, Firefox, Edge) for accessing the Streamlit dashboard.
- Network: Internet connection required for initial model download from Hugging Face Hub.

2.5 Design and Implementation Constraints

- The system is constrained to use the cardiffnlp/twitter-roberta-base-sentiment-latest model to ensure reproducibility.
- GPU VRAM limitations (4 GB) impose a batch size constraint of 8 during inference.
- Model evaluation is limited to a maximum of 1,000 comments to remain within VRAM bounds.
- The dataset must be pre-placed in the data/ directory in CSV format prior to application launch.
- The application is designed for local or Streamlit Cloud deployment; it does not include production-grade authentication or database integration.

2.6 Assumptions and Dependencies

- The YouTube Comments Sentiment Dataset (youtube_comments_cleaned.csv) is assumed to be available and pre-labeled with Positive, Negative, and Neutral categories.
- CUDA drivers and PyTorch with CUDA support are assumed to be installed for GPU operation.
- All Python dependencies listed in requirements.txt are assumed installable via pip.
- The Hugging Face model repository remains publicly accessible for initial download.

3. Specific Functional Requirements

3.1 Data Loading and Management

FR-1.1: Dataset Loading

The system shall load the YouTube comments dataset from `data/youtube_comments_cleaned.csv` upon application startup. The dataset must contain at minimum a `CommentText` column and a `Sentiment` column with values Positive, Negative, or Neutral.

FR-1.2: Sample Configuration

The system shall provide a sidebar slider allowing the user to select a sample size between 2,000 and 40,000 comments in increments of 1,000. The selected sample shall be drawn randomly with a fixed random seed (42) for reproducibility.

FR-1.3: Data Caching

The system shall cache loaded datasets and models using Streamlit's `@st.cache_data` and `@st.cache_resource` decorators to prevent redundant I/O operations and model reloading across user interactions.

3.2 Text Preprocessing

FR-2.1: Text Cleaning

The system shall apply an automated text cleaning pipeline to all comments prior to analysis. The pipeline shall include: conversion to lowercase, removal of URLs and HTML tags, removal of special characters and punctuation, tokenization using NLTK, and removal of English stop-words.

FR-2.2: Processing Time Tracking

The system shall measure and display the total processing time (in seconds) for the complete preprocessing and analysis pipeline, including the device name (GPU or CPU) on which the inference was performed.

3.3 Sentiment Classification

FR-3.1: Model Loading

The system shall load the pre-trained RoBERTa model (`cardiffnlp/twitter-roberta-base-sentiment-latest`) from Hugging Face Transformers. The model shall be loaded onto the GPU if a CUDA-compatible device is available; otherwise it shall fall back to CPU.

FR-3.2: Batch Inference

The system shall perform batch sentiment inference with a configurable batch size of 8 on the evaluation subset. Each comment shall be truncated to a maximum token length of 512. The system shall return probability scores for all three sentiment classes.

FR-3.3: Label Mapping

The system shall map raw model output labels (positive, negative, neutral) to standardized class labels (Positive, Negative, Neutral) regardless of case. Any unmapped label shall default to Neutral.

FR-3.4: Model Evaluation Toggle

The system shall provide a checkbox in the sidebar labeled 'Run BERT Model Evaluation'. When enabled, the system shall evaluate the model on up to 1,000 comments from the sampled dataset. When disabled, the system shall use a fixed sample of 8,000 comments for KPI computation only, without model inference.

3.4 KPI Dashboard

FR-4.1: Summary Metrics

The system shall display the following KPIs in a four-column layout at the top of the dashboard:

- Total Comments Analyzed (integer count with comma formatting).
- Positive Sentiment Percentage (one decimal place, e.g., 54.3%).
- Negative Sentiment Percentage (one decimal place).
- Neutral Sentiment Percentage (one decimal place).

FR-4.2: Model Performance Metrics

When model evaluation is enabled, the system shall compute and display the following metrics using scikit-learn:

- Overall Accuracy (percentage).
- Weighted Precision (percentage).
- Weighted Recall (percentage).
- Weighted F1-Score (percentage).
- Per-class classification report in tabular format (Precision, Recall, F1-Score, Support per class).

3.5 Keyword Analysis and Visualization

FR-5.1: Overall Keyword Frequency

The system shall compute and display the top 20 most frequent keywords across all comments in a horizontal bar chart using Streamlit's native chart component.

FR-5.2: Per-Sentiment Keyword Frequency

The system shall compute and display the top 15 most frequent keywords separately for each of the three sentiment classes (Positive, Negative, Neutral) as individual bar charts organized in tabs.

FR-5.3: Word Cloud Generation

The system shall generate and render word cloud visualizations for each sentiment class using the WordCloud library. Word clouds shall be generated with a width of 800px, height of 400px, white background, and a maximum of 150 words. The three word clouds shall be displayed side-by-side in a three-column layout.

3.6 Live Prediction

FR-6.1: User Input Interface

The system shall provide a text area on the dashboard where users can paste one or more YouTube comments, one per line, with a display height of 150px.

FR-6.2: Real-Time Classification

Upon clicking the 'Analyze with BERT' button, the system shall split the input by line, pass all non-empty comments to the RoBERTa classifier, and display the predicted sentiment label with confidence score and a color-coded emoji indicator for each comment:

- Green circle (🟢) for Positive sentiment.
- Red circle (🔴) for Negative sentiment.
- White circle (⬜) for Neutral sentiment.

FR-6.3: Spinner Feedback

The system shall display a loading spinner with the active device name while inference is in progress.

4. Non-Functional Requirements

4.1 Performance Requirements

ID	Requirement	Metric	Target
NFR-1	Inference Speed (GPU)	Comments/second	≥ 50 comments/sec on NVIDIA GTX 1650 (4GB)
NFR-2	Inference Speed (CPU)	Comments/second	≥ 5 comments/sec on standard CPU
NFR-3	Dashboard Load Time	Seconds	< 10 seconds for initial load with model caching
NFR-4	Live Prediction Latency	Seconds per request	< 3 seconds for up to 10 user-provided comments (GPU)
NFR-5	Dataset Processing	Seconds	< 60 seconds for preprocessing 40,000 comments

4.2 Reliability and Availability

- The system shall maintain stable operation during continuous sessions of up to 4 hours without memory leaks or application crashes.
- The system shall gracefully handle GPU unavailability by automatically falling back to CPU inference without user intervention.
- The system shall handle missing or malformed rows in the dataset by skipping invalid entries without terminating the application.

4.3 Usability Requirements

- The dashboard shall be navigable by a non-technical user within 5 minutes of first use without any training or documentation.
- All KPI metrics shall be clearly labeled with units and descriptions.
- The sidebar settings shall be logically organized with descriptive headers.
- Color-coded sentiment indicators shall adhere to intuitive conventions (green = positive, red = negative, neutral = white/grey).

4.4 Security Requirements

- The application does not handle personally identifiable information (PII) and therefore does not require user authentication.
- The dataset file shall be stored locally and not transmitted to any external server.
- Model inference is performed locally; no comment data is sent to third-party APIs during prediction.

4.5 Maintainability Requirements

- The codebase shall be modularized into at least two Python files: app.py (UI and orchestration) and utils.py (preprocessing and keyword utilities).
- All functions shall include descriptive naming following Python PEP 8 conventions.
- Dependency versions shall be explicitly pinned in requirements.txt to ensure reproducible installations.

4.6 Portability Requirements

- The application shall run on any system supporting Python 3.9+ and pip package installation.
- The system shall be deployable to Streamlit Cloud without modification to the application code.
- No platform-specific file paths shall be hard-coded; all paths shall use relative references.

5. System Architecture and Models

5.1 System Architecture Overview

The YT-Sentiment system follows a three-layer architecture:

- Data Layer: CSV dataset storage, NLTK preprocessing, and data caching via Streamlit.
- Model Layer: Hugging Face Transformers pipeline hosting the RoBERTa model, PyTorch inference engine with CUDA support.
- Presentation Layer: Streamlit web application providing the interactive dashboard, KPIs, visualizations, and live prediction interface.

5.2 Technology Stack

Layer	Technology	Purpose
Frontend / UI	Streamlit	Interactive web dashboard
Model Inference	Hugging Face Transformers + PyTorch	RoBERTa sentiment classification
Preprocessing	NLTK, pandas	Text cleaning and dataset management
Visualization	Matplotlib, Seaborn, WordCloud	Charts and word cloud generation
Evaluation	scikit-learn	Accuracy, Precision, Recall, F1-Score
GPU Acceleration	PyTorch CUDA (11.8)	Accelerated model inference

5.3 Data Flow

The following sequence describes the primary data flow through the system:

9. User launches the Streamlit application (`streamlit run app.py`).
10. Application loads the CSV dataset and applies text preprocessing via `utils.clean_text()`.
11. The RoBERTa model is loaded from Hugging Face (cached after first download).
12. If evaluation mode is active, the model performs batch inference on up to 1,000 comments.
13. KPI metrics are computed and displayed in the dashboard header.
14. Keyword frequencies are computed per class and rendered as bar charts.
15. Word clouds are generated and displayed in the visualization tab.
16. User may optionally paste comments into the live prediction panel and click Analyze.
17. The model classifies each pasted comment and returns labeled results with confidence scores.

5.4 Module Descriptions

5.4.1 `app.py` — Main Application Module

Serves as the entry point and orchestrator for the Streamlit application. Responsibilities include: GPU/CPU device detection, model loading with caching, dataset loading and sampling, preprocessing invocation, KPI computation and rendering, visualization rendering, and live prediction interface management.

5.4.2 `utils.py` — Utility Module

Contains reusable helper functions for text processing and keyword extraction. Key functions include: `clean_text()` for text normalization and stop-word removal, `get_top_keywords()` for computing overall term frequencies, and `get_top_keywords_per_sentiment()` for per-class keyword analysis.

6. Constraints and Limitations

6.1 Hardware Constraints

- GPU VRAM is constrained to 4 GB, limiting inference batch size to 8 and evaluation subset to 1,000 comments per run.
- CPU inference is approximately 10x slower than GPU; large sample evaluations are not recommended in CPU mode.

6.2 Software Constraints

- The system is constrained to using the cardiffnlp/twitter-roberta-base-sentiment-latest model; alternative models require code modifications.
- The dataset must contain CommentText and Sentiment columns with exact column name matching.
- Streamlit's execution model re-runs the entire script on each user interaction, which is mitigated via caching but limits stateful processing.

6.3 Academic Scope Limitations

- The system is an academic proof-of-concept and does not include user authentication, persistent storage, or enterprise-grade scalability.
- The sentiment labels in the dataset are taken as ground truth without re-verification or human annotation review.
- The system does not support multilingual comments; it is optimized for English-language text only.

Appendix A: Requirements Traceability Matrix

The following table maps each functional requirement to its corresponding system component:

Req. ID	Description	Module	Priority
FR-1.1	Dataset Loading	app.py	High
FR-1.2	Sample Configuration	app.py	Medium
FR-1.3	Data Caching	app.py	High
FR-2.1	Text Cleaning	utils.py	High
FR-2.2	Processing Time Tracking	app.py	Low
FR-3.1	Model Loading	app.py	High
FR-3.2	Batch Inference	app.py	High
FR-3.3	Label Mapping	app.py	High
FR-3.4	Evaluation Toggle	app.py	Medium
FR-4.1	Summary KPIs	app.py	High
FR-4.2	Model Performance Metrics	app.py	High
FR-5.1	Overall Keywords	utils.py, app.py	Medium
FR-5.2	Per-Sentiment Keywords	utils.py, app.py	Medium
FR-5.3	Word Clouds	app.py	Medium
FR-6.1	User Input Interface	app.py	High
FR-6.2	Real-Time Classification	app.py	High
FR-6.3	Spinner Feedback	app.py	Low

Appendix B: Revision History

Version	Date	Description	Author
1.0	May 2026	Initial SRS document created for academic submission.	Team (Waleed, Hanzla, Taha, Ali)